

Code-Layout, Readability and Best Practices

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Project Folder Layout

The development phase code is organized by Epic, mirroring the project's task breakdown, so any contributor can quickly locate the relevant script or notebook.

- Epic 1 - Data Collection and Architecture Design/ - dataset + architecture notes
- Epic 2 - Visualizing and Analysing the Data/ - EDA.ipynb
- Epic 3 - Data Pre-processing/ - preprocessing.py
- Epic 4 - Model Building/ - model_building.py
- Epic 5 - Application Building/ - app.py, templates/, static/

Coding Standards Followed

- PEP 8 style guide for Python code (naming, spacing, line length).
- Functions kept small and single-purpose (e.g. handle_missing_values, scale_data, train_xgboost).
- Docstrings added to every function explaining its role in the pipeline.
- Consistent naming convention: snake_case for variables/functions.
- Configuration values (test size, random state) kept as named parameters with sensible defaults.

Readability Practices

- Each pipeline stage (preprocessing, model building) lives in its own file rather than one large script.
- Comments map each function back to its corresponding user story (e.g. Story: Balancing the Dataset).
- Notebooks use markdown headers to separate Univariate, Bivariate, and Multivariate analysis sections.

Version Control Practices

- Git used for tracking changes, with descriptive commit messages.
- .gitignore excludes generated artifacts (model.pkl, __pycache__, large CSVs) to keep the repo clean.